

There will be three parts or modules in this project with specific working and role

Description of modules (Role in project):

Authentication:

Will do tasks like encryption /decryption basically security stuff

Hangman Logic:

Game logic will contain string manipulation

Results compilation:

Will handle numeric manipulation

Working summary:

Authentication:

Will load encrypted password that is stored in a file,

Then we will input password from the user

Then we decrypt the password from the file and then compare both of decrypted and input password and decide accordingly.

Hangman Logic:

We will load a word with different missing letters , the user has to guess these letters in given number of tries.

And we will manage score as well , for each correct guess , we will add 3 marks to the score and for each wrong guess we will deduct 1 mark from the score.

And if no wrong guess was made the score should be multiplied by 3

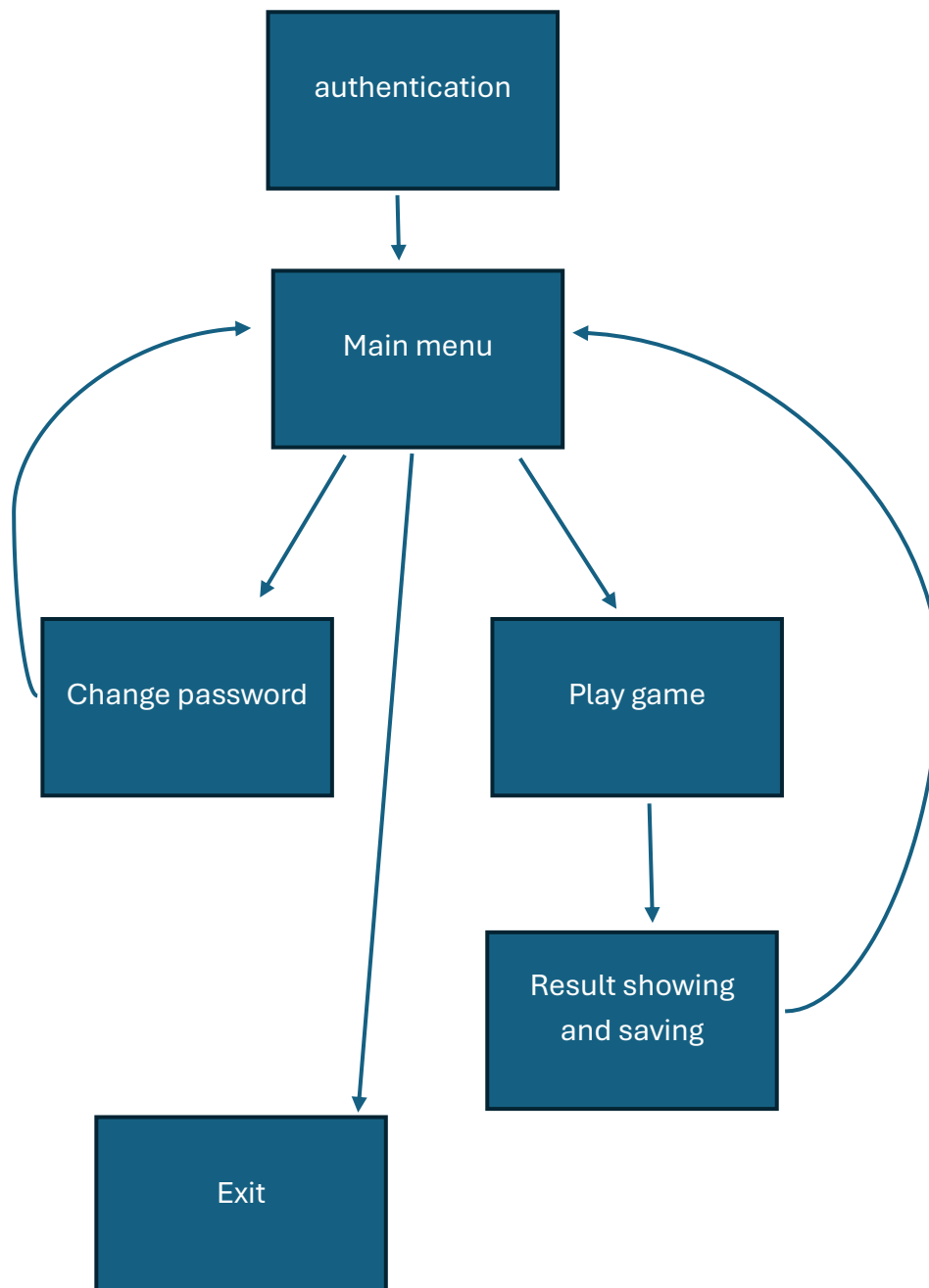
Results :

Will show current score after full round

And save the average score and no. of rounds played

Check if the new score is the highest, if yes then store that score in the max variable , else do nothing

Flow diagram :



Complete pseudo code:

Authentication:

Enc_pass = load from file (filename)

Saved_password = decrypt (Enc_pass)

Loop pass :

 Input_pass = input password from user

 If input_pass == saved password

 // make some global variables for results

 Highest_score = 0, no_of_rounds_played = 0 , avg_score = 0

 Go to main menu

 Else :

 Loop pass

Main menu :

// it will be a simple menu with 3 options ,

Enter 1 if you want to change password , Enter 2 if you want to play game , enter anything else to quit the game

Op = input from user option

If op == 1

 Go to change password()

Else if op == 2 go to hangman ()

Else quit

change password :

new_pass = input new password from user

// password should be exactly 6 characters long , no less , no more

If new_pass is valid

Save the new password in the file

Else print (invalid password entered)

Go to main menu

Hangman :

// we will make three levels hardcoded with 3 strings

Lv1_string = “something”

Lv2_string = “something”

Lv3_string = “something”

Total score = 0

Score+= Start_level(Lv1_string)

Score+= Start_level(Lv2_string)

Score+= Start_level(Lv3_string)

// we have to implement Start_level as a macro to complete the rubrics of the assignment

Show total_score

Go to results (total_score)

Go to main menu

Start_level (levelString):

Wrong_guess_limit = somenumber

Print string

// the string will be displayed with random places replaced by '_'

// those letters will be stored in hidden_letters variable

hidden_letters = ''

No_mistake = 1

Loop play:

 C = read character from user

 If c in hidden letters

 Score += 2

 Print string , with the character c showing

 If hidden_letters == null // means the word is completely visible then

 Break

 If c is not in hidden_letters

 Score -= 1

 No_mistake = 0

 Wrong_guess_limit -= 1

 If wrong_guess_limit == 0

 Display (you failed)

 Return 0

If no_mistake == 1

Score *= 3

Return score

Results (new_score) :

display the new result

//Access the global variables made after authentication

If new_score > highest_score :

 Display (“ new highest score “)

 Highest_score = new_score

average_score =((average_score * no_of_rounds) + new_score) / (no_of_rounds +1)

display average _score

no_of_rounds += 1

return